Multiple choice (each question has only one correct answer). You don't need to show your work.

1. Let A be a matrix with 3 rows and 4 columns. If B is a matrix so that BA has 5 rows and 4 columns, then what must be the shape of B?

   a) 5×4.

   b) 3×3.

   c) 5×3.

   d) 3×5.

2. A system of 4 equations and 5 unknowns

   a) Always has no solution.

   b) Always has exactly one solution.

   c) Always has infinitely many solutions.

   d) None of the above.

3. Let A be a 3×3 matrix. Which of the following is true?

   a) If A is in reduced row echelon form, then it must be the identity matrix.

   b) If A is not invertible, then every equation $A\vec{x} = \vec{b}$ has either no solutions or infinitely many solutions.

   c) Both a) and b) are true.

   d) Neither a) nor b) is true.

Longer questions. Show all your work!

4. Let $A = \begin{bmatrix} 2 & a \\ b & -3 \end{bmatrix}$ be a matrix with unknown entries a and b, and suppose $A \begin{bmatrix} -1 \\ 5 \end{bmatrix} = \begin{bmatrix} 13 \\ -16 \end{bmatrix}$. Find a and b or explain why it is not possible to solve for their values.

5. A framing store offers various sizes of picture frames: 4-inch, 8-inch, and 12-inch. They only sell their frames in the following three bundles.

- Bundle A: one 4-inch frame and two 8-inch frames for \$8.
- Bundle B: one 4-inch frame and two 12-inch frames for \$13.
- Bundle C: two 4-inch frames, four 8-inch frames, and four 12-inch frames for \$36.

a) Write a formula for a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} \#\text{ 4-inch frames} \\ \#\text{ 8-inch frames} \\ \#\text{ 12-inch frames} \end{bmatrix},$$

where a, b, and c are the number of bundles A, B, and C purchased, respectively.

b) Suppose the store allows for refunds. Is it possible to buy and refund bundles to leave the store with a single 4-inch frame? If so, how much will it cost?

6. (12 points) Consider the system of linear equations

$$\begin{aligned} 2a - 3b + c &= 2 \\ -a + 2b - 4c + 2d &= -3 \\ a - b - 3c + 2d &= -1. \end{aligned}$$

a) Write the system as an augmented matrix and solve it for a, b, c, and d if possible.

b) Let $T : \mathbb{R}^4 \to \mathbb{R}^3$, defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}\right) = \begin{bmatrix} 2a - 3b + c \\ -a + 2b - 4c + 2d \\ a - b - 3c + 2d \end{bmatrix},$$

be the linear transformation that sends a vector of variables to the left sides of the system in part a). Is T one-to-one? It is onto?

c) Find a basis for $\ker T$ and a basis for image T.

7. (12 points) Let $A = \begin{bmatrix} 3 & 4 & 1 \\ -2 & -3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and let T be the linear transformation corresponding to A.

a) Find a formula for T^{-1}.

b) Explain with a sentence why the columns $\vec{v}_1 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ of A form a basis for $\mathbb{R}^3$, and express the vector $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ in this basis.

c) The linear transformation $S : \mathbb{R}^2 \to \mathbb{R}^3$ satisfies $S\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ and $S\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$. Find the matrix corresponding to $T \circ S$.